Round 4: follow-up experiments on items 1–3 (Addendum H)

Status. Nine follow-up experiments (H1–H9) were frozen in `PROTOCOL.md` (Addendum H, `f3bcf9a63`) before any of them ran. Amendments H-a and H-b were committed before the runs they govern. Seven experiments ran and are recorded:

- H1, H2 (with H3 folded in) and H6 on RunPod CPU pods;
- H4, H5 and H7 in this session's container.

H8 was not run, and H9 is answered by existing studies. Every cost is a counted operation. Every discrete log was verified (`k·P = Q`): 3,860 in total.

Where it ran. The repo's `ephemeral` runner could not authenticate (the container has an AWS access-key ID but no secret). Jobs went instead through `runpod/podrun.py`. A pod fetches the tool at a pinned, pushed commit from the public repository, checks each file's SHA-256, runs the command fixed at launch, and serves only read-only results. Nothing is uploaded into the pod. Every pod was terminated after its fetch, and each run directory keeps `pod.json`, `host.txt` and `rusage.txt`.

Answer

Item 3: `n = 2` through genus-2 Jacobians.

1. **The route reaches prime-order curves [E].** Any `E/F_{p²}` with `j ∉ F_p` is glued to `E^σ` along `ψ(a) = a^{p³}` on the 2-torsion roots. This gives a genus-2 `C/F_p` with `Jac(C) ~ Res E` and an explicit transfer.
   - It descends whenever the 2-torsion cubic is irreducible or split over `F_{p²}`.
   - With exactly one rational 2-torsion point, no Galois-equivariant gluing satisfies the descent cocycle. So `h ≡ 2 (mod 4)` is not reached, which matches `4 | #E` throughout the Scholten family.
   - Checks G1–G5 hold on every instance (see the checks table).
2. **On prime-order `E` (`h = 1`)** the first-cycle IC needs **0.43–0.44·p** Jacobian operations, against rho's 0.78–0.96·p: about half.
3. **Per-operation cost.** The reduced sextic-model formulas cost 41 M + 1 I per addition and 51 M + 1 I per doubling, against 12 M + 1 I for an `E(F_{p²})` addition.
4. **Ranking at `h = 1`.** With the split test priced honestly (one Legendre symbol at one `I`), the pricing band is **0.59–1.94 (unresolved) at every size from 28 to 40 bits**, with the IC cheaper in up to 6 of 12 cells.
   - Distance to rho at measured cost (`I = 3M`): 1.43–1.81×.
   - Break-even: a genus-2 addition of about 22 M + I. That is the best published figure, not yet measured here.
5. **The cofactor law (H1) closes the large-h hope.**
   - Arm J's trials scale as `h^{−0.402}` (95% [0.380, 0.424]); rho's as `h^{−0.468}`, near the theoretical ½.
   - Arm J's count advantage is largest at small `h` and is gone by `h ≈ 16`.
   - The frozen decision rule applies: on large-cofactor curves (FourQ-like `h = 392`) this route never beats rho on count.
6. **Larger sizes (H6, 44–48 bits)** reproduce the constants.
   - Rho: `steps/p` 0.33 and 0.45, against theory 0.31 and 0.44.
   - Arm J: `steps/p` 0.30 and 0.31, band unresolved.
   - Arm D (decomposition): 28–87× rho.
7. **Memory (H4) is the real limit.** Time × memory does not stay near `p²/9`, as predicted: it falls from 0.040·p² to 0.004·p² as θ goes from ½ to ⅛. With few stored relations, the walk solves mostly through repeated relations, which is distinguished-point rho inside the IC walk. A memory-capped Arm J turns back into rho, so the IC's constant-factor count advantage exists only with Θ(p) memory.
8. **Automorphisms (H5) [E].** An exact census over `p = 5…13` finds 6, 6, 14 and 14 classes that are simple over `F_p`, Res-shaped, and have automorphisms beyond `±1` (orders 4 to 120). The total mass equals `p³` exactly at every `p`. These are candidates for an IC gain of `k` against rho's √k. Measuring that needs such curves at cryptographic size and a rho that uses the same automorphism; not done.

Item 1: decomposition over `F_{q^n}`.

9. **Coset confinement measured on the target subgroup (H7) [E].** This is coverage of the ℓ-subgroup itself, not the whole group.
   - **Binary `n = 2`:** subfield over null gives a ratio of **1.86 ± 0.16** (21 curves). That confirms the corrected reading of round 3: the subfield factor base reaches 0.75 of `2E` and of each deeper `2^k E`, against a Poisson value of 0.39.
   - **Odd characteristic, size-matched split twists (`E ≅ Y² = Π(X − d e_i)`, `e_i ∈ F_q`, `d` a square), `n = 2`:** ratio **2.39 ± 0.21** (12 curves). The subfield factor base covers the whole ℓ-subgroup (1.0, against 0.27–0.71 for null).
   - **Generic odd curves:** 1.17 ± 0.10, near the no-effect value.
   - **`n = 4` cells:** too small to read (s.e. 0.13–0.71).
   - The gain is bounded by the cofactor's 2-part. It is a cost-model fact about these curve families, not a new algorithm, and prime-order curves gain nothing.

Item 2: prime fields.

10. **H9 needs no new run.** The PKM solving degree is the subject of `research/notes/index-calculus/RESEARCH_PKM_TOWER_ORACLE.md`, rounds 1–7 and later studies (for example `D = 7` at `m = 3`, `N = 18`, and at `m = 4`, `N = 16`). Re-measuring would duplicate them; this round cites them. Round 3 settled coverage: no effect.

Not run: H8 (decomposition crossover at `n = 3`). The repo's Semaev tools are binary-field only. A descended-`S_4` solver over `F_{p³}` (Gröbner or resultant) is a separate build. It is the next item-1 experiment.

H1: the cofactor law (run `H1-hsweep-run-20261018T2240Z`, RunPod, 16 vCPU)

Scale. The Scholten family, with `h ∈ {4, 8, 12, 16, 24, 32, 48, 64, 96, 128}` × `log2 #E ∈ {32, 36, 40}`. Each cell has 4 instances × 10 targets, with arms R and J, plus Jind at 32 bits. All answers verified.

Fits. On cell means, with a 2,000-sample bootstrap over instances:

arm	α (trials/p ∝ h^−α)	95% interval
J (walk)	0.402	[0.380, 0.424]
Jind (independent trials, 32 bits)	0.368	[0.334, 0.400]
R (rho)	0.468	[0.446, 0.490]

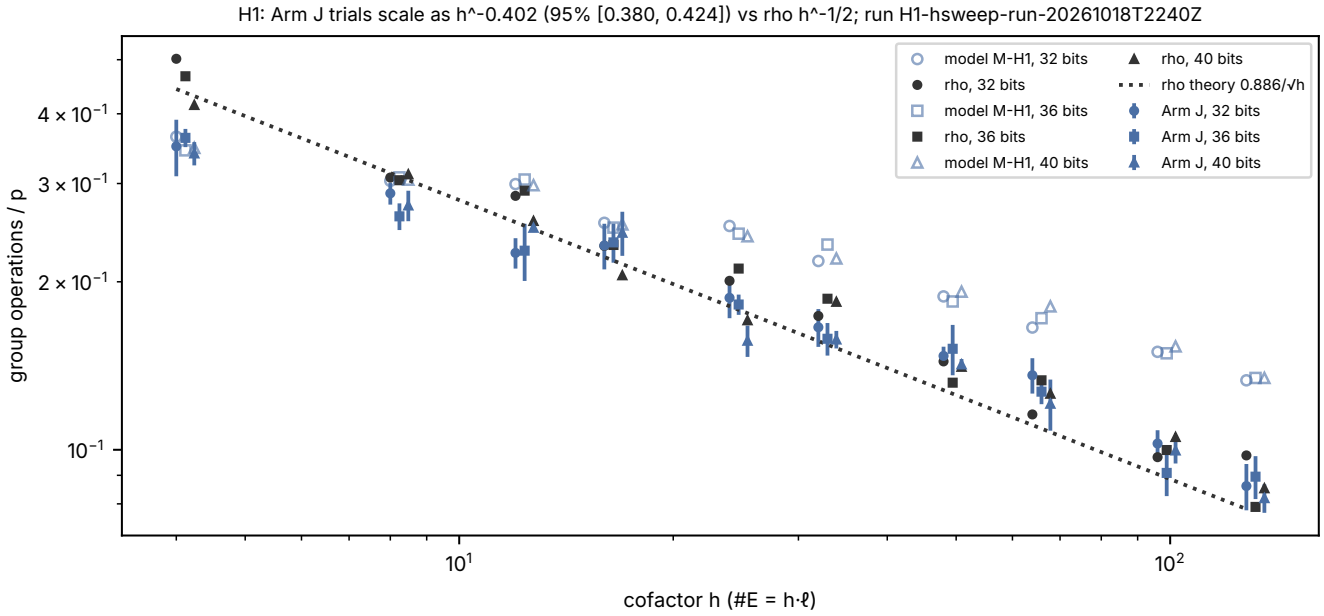


Figure 1: H1

Model M-H1. This is the uniform admissible-pair graph on the instance's real degree-1 points and their cofactor classes, with sign parity. It matches the measured trials within about 10% at `h ≤ 16`. At `h = 64–128` it overpredicts by 1.3–1.6×. Its uniform-pair assumption does not hold there: relations are not uniform over admissible pairs once the classes are small. The measured law is the result; the model is diagnostic only.

H2/H3: the general gluing (run `H2-glue-run-20261018T2240Z`, RunPod, 16 vCPU)

Construction.

- Solve `R ab + S a + T b = 1` over the étale algebra `F_{p²}[z]/(g)`, with `b = ψ(a) = z^{p³}`. This gives `s_2 = −T/R`, `t_2 = −S/R`, `k = s_2t_2 + 1/R` and `s_1 = k·g(s_2)`.
- That defines `C: y² = g(s_1x² + s_2)/s_1²` over `F_{p²}`.
- Descent datum: `x ↦ λ/x^σ, λ² = k/N(s_1)`.
- The `F_p` model comes from a Baer-subline cross-ratio map. A rational point is moved to infinity, so the model has two rational points at infinity. It is twisted when needed, so that `τ` is nonzero on the `ℓ`-part.
- The IC columns are `x_P = [2P − D∞]`, so a relation reads `(x_P + x_Q)/2`.

Checks. On 16 instances plus G5:

check	result
`t_2 = σ(s_2), λ² ∈ F_p`	16/16
G1: `#C(F_p) = p + 1; #C(F_{p²}) = p² + 1 − 2t` (`p < 400`)	16/16; 8/8
G2: associativity, inverse, `N·A = 0`	all true; `N·A` "degenerate" twice at `p ≤ 277`, never false
G3: homomorphism, `τ ≠ 0, ℓ·τ = 0`	all true; one "degenerate" at `p = 257`
G5: reduced formulas = generic law	2,400/2,400 pairs

Measured (`log2 #E` 28–40; 4 instances × 10 targets per cell; all answers verified). Group operations per `p`, mean ± s.e.:

bits	h	Arm J	rho	Arm J cost / rho, pricing band	verdict
28	1	0.435 ± 0.017	0.956 ± 0.064	0.59–1.51	unresolved
32	1	0.426 ± 0.018	0.844 ± 0.069	0.71–1.71	unresolved
36	1	0.439 ± 0.014	0.778 ± 0.076	0.82–1.94	unresolved
40	1	0.443 ± 0.016	0.860 ± 0.068	0.76–1.80	unresolved
36	4	0.337 ± 0.018	0.403 ± 0.041	1.18–2.79	rho cheaper
40	4	0.350 ± 0.018	0.445 ± 0.031	1.14–2.69	rho cheaper

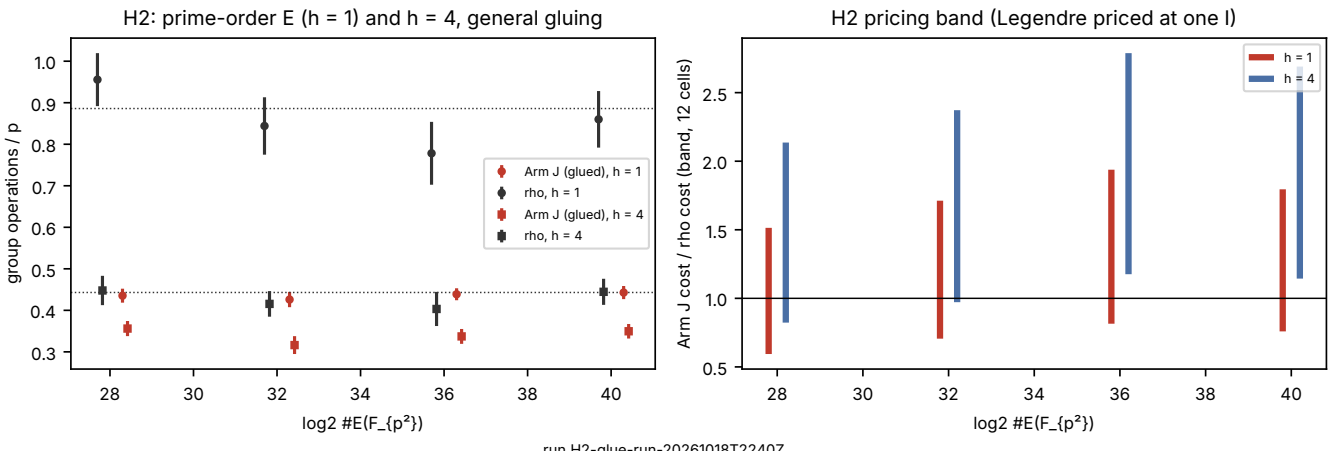


Figure 2: H2

Distance to rho (prime-order `E/F_{p²}`). About 1.7× at 40 bits under measured costs. Under the IC-favourable price (one Jacobian operation = one `E(F_{p²})` addition), the IC is cheaper: 0.76× at 40 bits. Closing the gap needs a genus-2 sextic-model addition of about 22 M + I, measured. This round's reduced formulas take 41 M + I.

H4: memory against time (run `H4-memsweep-run-20261018T2209Z`)

bits	θ	trials/p	stored/p	time·memory/p²
32	½	0.563	0.0719	0.0405
32	¼	0.624	0.0201	0.0125
32	⅛	0.700	0.0056	0.0039
36	½	0.498	0.0624	0.0311
36	¼	0.588	0.0182	0.0107
36	⅛	0.617	0.0048	0.0030

At θ = 1 (`h = 4`) the trials are about 0.35·p, with about 0.18·p stored.

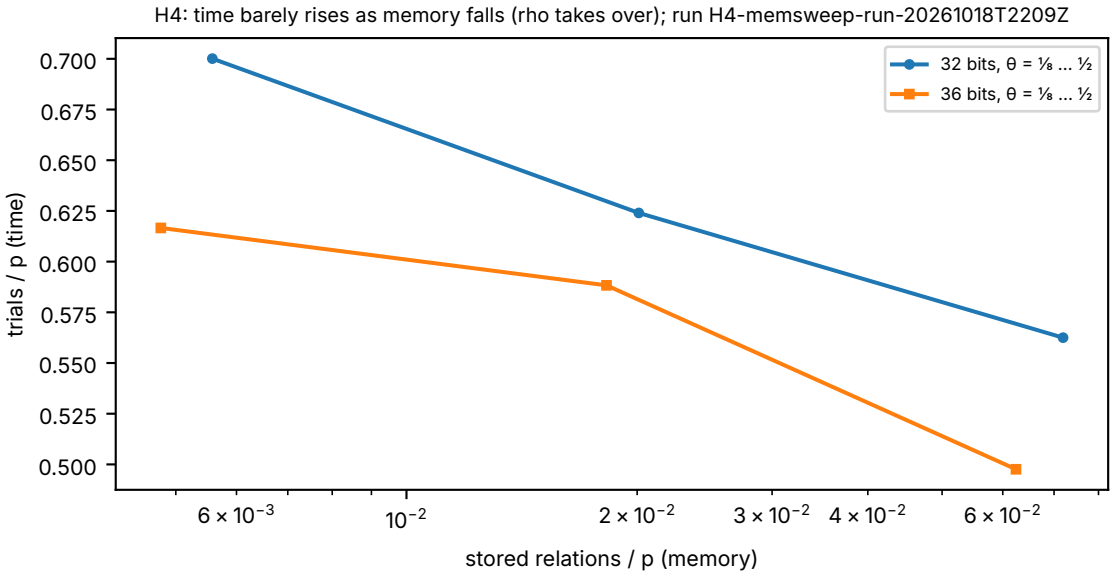


Figure 3: H4

The prediction (time ∝ 1/θ) is falsified. As θ falls, the answers come from relations the walk repeats, so the arm becomes rho with distinguished points (H4 used the walk arm; a Jind-θ arm would isolate the graph mechanism, and was not run). The IC's count advantage at θ = 1 depends on Θ(p) relation storage. At cryptographic sizes that is the binding constraint, and the comparison should be read as area-time.

H5: automorphisms (run `H5-aut-census-run-20261018T2227Z`) [E]

Classes simple over `F_p` with even L-polynomial (`a_1 = 0`), by `|Aut|`, which includes the hyperelliptic involution:

p	mass Σ1/|Aut| (= p³)	|Aut| = 2	4	6	8	12	24	120	with |Aut| > 2
5	125	31	1	4	–	–	–	1	6
7	343	85	3	2	–	–	1	–	6

11	1331	306	3	9	1	1	–	–	14
13	2197	450	6	7	–	–	1	–	14

So Res-shaped, simple Jacobians with extra automorphisms over F_p exist. Whether the IC's gain of k beats rho's √k on them needs the same automorphism granted to rho in Jac(C), and needs a family at large p. Not measured.

H6: larger sizes (run H6-large-run-20261018T2210Z, RunPod, 16 vCPU)

Arms R, J and D at c = 2; 3 instances × 10 targets per size; all 180 answers verified.

bits	rho steps/p	theory	Arm J steps/p	Arm J band	Arm D band
44	0.328 ± 0.026	0.313	0.295	0.99–2.96, unresolved	28.4–80.9, rho cheaper
48	0.448 ± 0.042	0.443	0.311	0.77–2.32, unresolved	28.1–86.8, rho cheaper

H7: coverage of the target subgroup (run H7-subgroup-run-20261018T2209Z) [E]

Ratio of ℓ-subgroup coverage, subfield over null; mean ± s.e. over curves:

cell	n = 2	n = 4
binary (m = 4…10 / 3…5)	1.86 ± 0.16 (21)	2.07 ± 0.71 (9)
odd generic	1.17 ± 0.10 (12)	1.33 ± 0.53 (9)
odd split twist, frozen (d generic)	1.18 ± 0.55 (12)	0.40 ± 0.15 (9)
odd split twist, size-matched (d square)	**2.39 ± 0.21** (12)	0.74 ± 0.13 (9)



Figure 4: H7

Why the frozen split-twist cell is noisy. With d a non-square, the factor base is empty apart from the 2-torsion points (Amendment H-a).

Why the n = 4 cells are unreadable. Their ℓ-subgroups are small (tens to hundreds of points).

Mechanism.

- **Binary:** Tr(x) = 0 for x ∈ F_q.
- **Odd split twists:** every X − d e_i ∈ d·F_q has the same quadratic character.

In both cases the factor base sits in one coset of 2E, and its sums land in 2E, which contains every target.

Checks

check	result
every discrete log verified (k·P = Q)	H1 2,800; H2 640; H4 240; H6 180 (3,860 in this round, all verified)
rho positive control (K6) at 44/48 bits	matches 0.886·√ℓ/p within 1 s.e.
G1–G5 (gluing)	see H2; no false check
H5 mass Σ 1/|Aut| = p³	exact at p = 5, 7, 11, 13
unit tests	n2_routes 5, decomposition_coverage 5, genus2_aut_census 1: pass

What is not established

- **H2.** The 22 M + I break-even figure is an estimate. No formula at that cost was implemented here.
- **H4.** The memory result uses the walk arm.
- **H5.** No measurement beyond the census.
- **H7.** The n = 4 cells are underpowered.

- **H8** was not run.
- **Novelty.** None of these results has a novelty audit. The general gluing is very likely Scholten's (2003) construction, written down here with the `a^{p³}` cocycle made explicit. The statement that `h ≡ 2 (mod 4)` is unreachable needs checking against that paper.

Next improvements

1. **Measure a cheaper sextic-model addition.** This decides `h = 1`; the target is 22 M + I or less. Use the explicit real-model formulas, with Karatsuba-style products and a shared inversion, and verify them against the generic law.
2. **Make memory part of the comparison.** Report area-time for every Arm J row, and test a Jind-θ arm that isolates the graph from the walk.
3. **H5 follow-up.** Build a large-`p` family of Res-shaped curves with an order-4 automorphism. Measure Arm J on the automorphism-reduced graph against rho in `Jac(C)` with the same automorphism.
4. **H8.** Build a descended-`S_4` oracle over `F_{p³}` and measure the `n = 3` crossover.
5. **H7.** Enlarge the binary and odd `n = 4` cells to readable ℓ-subgroup sizes.

Run reports

- `reports/H1-hsweep-run-20261018T2240Z/run-report.pdf`
- `reports/H2-glue-run-20261018T2240Z/run-report.pdf`
- `reports/H4-memsweep-run-20261018T2209Z/run-report.pdf`
- `reports/H5-aut-census-run-20261018T2227Z/run-report.pdf`
- `reports/H6-large-run-20261018T2210Z/run-report.pdf`
- `reports/H7-subgroup-run-20261018T2209Z/run-report.pdf`